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$$\lim_{x\to+\infty}\left(x - x^2 \ln\left(1+\frac{1}{x}\right)\right)$$

Stel $t = \dfrac{1}{x}$, als $x \to +\infty$, dan $t \to 0$

$$\Rightarrow \lim_{t\to 0} \frac{1}{t} - \frac{1}{t^2}\ln(1+t) = \lim_{t\to 0}\frac{t-\ln(1+t)}{t^2}$$

$$= \frac{0}{0} \overset{H}{=} \lim_{t\to 0}\frac{1-\dfrac{1}{1+t}}{2t}$$

$$= \lim_{t\to 0}\frac{\dfrac{1+t-1}{1+t}}{2t}$$

$$= \lim_{t\to 0}\frac{t}{2t(1+t)} = \frac{1}{2}$$

References